

CERTIFICATE OF CALIBRATION

The under-mentioned item has been calibrated at the following points in the Michell Instruments Humidity Calibration Laboratory against test equipment traceable to the NATIONAL PHYSICAL LABORATORY, Middlesex, United Kingdom and to the NATIONAL INSTITUTE OF STANDARDS & TECHNOLOGY, Gaithersburg, Maryland, USA.
Dew point traceability to National Physical Laboratory: -90 to +90 °C
Dew point traceability to National Institute of Standards and Technology: -75 to +20 °C

Certificate Number	973726	Reference Number	AA14565
Test Date	31 March 2025	Test Equipment	Q0585
Sensor Serial No.	2C5B-030	Instrument Serial No.	Not Applicable
Model	EA2-TX-100		

The following calibration data was obtained by comparison with a Michell Instruments Precision Chilled Mirror Hygrometer:

Generated Dew point °C	Measured Dew point °C	Generated Dew point °C	Measured Dew point °C
-100.0	-100.0	-30.7	-30.7
-89.6	-89.6	-20.2	-20.2
-80.9	-80.9	-9.7	-9.7
-70.5	-70.5	0.5	0.5
-59.5	-59.4	10.4	10.5
-49.5	-49.5	18.6	18.6
-40.0	-39.9		

Comments:

Calibration PASS. The results at the tested dew points are within the sensor accuracy specification stated below.

Calibration Work Instruction used: 274

Sensor Accuracy: +/- 2 °C from -100 °C to +20 °C DP.

Dew point measurement uncertainties

Measurement (°C dew point)	Expanded Uncertainty (°C dew point)
+20	±0.40
-60	±0.60
-80	±0.90
-90	±1.20
-100 (not traceable)	±1.80

Uncertainty is linear between stated measurement points.

The uncertainties are based on a standard uncertainty multiplied by a coverage factor $k=2$, providing a level of confidence of approximately 95%.

Checked By

03 April 2025

Donna Johnson

Michell Instruments Ltd.
www.ProcessSensing.com
uk.michell.info@ProcessSensing.com